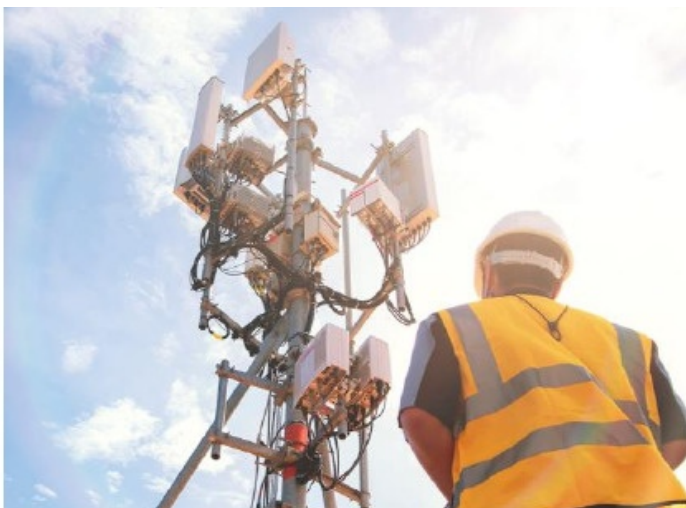


Model 12100 Series RF Power Sensors



Features

- Converge from 9kHz - 75GHz
- Accurate power measurements on any modulated signal
- Variety of options such as triggering, security, and analog outputs
- Command compatible with U2000
- SPI and I2C interfaces
- USBTMC, USBHID IVI foundation, and SCPI command set
- Extensive internal memory for trace and long term logging
- Real time clock and backup

Applications

- ATE
- Direct Control
- Research and Development
- Manufacturing
- Radar, satellite, and telecommunications

Model 12100 Series

9 kHz to 75 GHz RF Power Sensor



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GENERAL DESCRIPTION

Berkeley Nucleonics is excited to release our newest line of RF average power sensors: the Model 12100 Series. These versatile and easy-to-use sensors boast an impressive 9 kHz to 67 GHz bandwidth. With a dynamic range of -60 dBm to 26 dBm, these sensors can measure even the most subtle RF signals with industry-leading accuracy. This makes the Model 12100 an ideal choice for any RF application where obtaining true average power readings are critical for project success. Measuring average power on any type of modulation (802.11ac, gsm, cdma 2000, DCMA (IS-95), TDMA, Multi-Tone, OFDM, CW, QAM, etc) has never been easier. Since 1963 Berkeley Nucleonics has provided the highest standard of test and measurement equipment and the Model 12100 RF average power sensor is no exception.

These sensors offer simple, plug and play operation. This means no zeroing, no calibration before use, and no bulky meters and cable harnesses to manage. The Model 12100 sensors can be used with minimal setup time and hassle, saving valuable hours and allowing users to take measurements almost immediately whether they are in a lab or in the field. With a deep internal memory and internal or external triggering, long term data logging is managed elegantly. While at the forefront of RF power measurement technology, our innovative technology will not destroy your budget. We leverage modern components and chip sets which drive down cost. Our new sensor family is compact and rugged, unmatched in traceable power accuracy and yet costeffective for the budget constraints of our time. Visit the product page for a growing range of options addressing programmatic and embedded applications, excellent SCPI options, a real time clock and more.

BNC PowerEye is an included software package giving users a suite of handy tools. The software is easy to install, straightforward to use, and able to deliver information vital to RF engineers. A demo version is available online at www.berkeleynucleonics.com and software tools are always free. Whether you have needs for large system integration and user specified data inputs/outputs, or smaller, on-the-bench R&D projects, the Model 12100 sensor deserves your consideration. A quick phone call, email or live 'engineer-chat' will quickly illustrate why the T&M industry has turned to Berkeley Nucleonics routinely for over 60 years. We look forward to hearing from you.

SPECIFICATIONS

Frequency Range	12108L:	9 kHz to 8 GHz
	12108S:	1 MHz to 8 GHz
	12118L:	9 kHz to 18 GHz
	12118S:	1 MHz to 18 GHz
	12126L:	9 kHz to 26.5 GHz
	12126S:	1 MHz to 26.5 GHz
	12140L:	9 kHz to 40 GHz
	12140S:	1 MHz to 40 GHz
	12144L:	9 kHz to 44 GHz
	12144S:	1 MHz to 44 GHz
	12150L:	9 kHz to 50 GHz
	12150S:	1 MHz to 50 GHz
	12154L:	9 kHz to 54 GHz
	12167L:	9 kHz to 67 GHz
	12175:	50 GHz to 75 GHz
Input Power	All except 12175	-60 to +26 dBm
	12175:	-50 to +18 dBm
Continuous Average	Standard	+26 dBm (400 mW)
	12167L:	+18 dBm (63 mW)
	12175:	+18 dBm (63 mW)
Damage Level	Standard	+29 dBm (800 mW)
	12167L:	+23 dBm (200 mW)
	12175:	+23 dBm (200 mW)
Peak Power	Standard	+23 dBm (200 mW)
	12167L:	+26 dBm (400 mW)
	12175:	+18 dBm (63 mW)
	Damage Level	36 dBm (4 W); 12175: +23 dBm (200 mW)
Energy Per Pulse	Standard	20 W-us
	12167L:	5 W-us
	Damage Level	40 W-us
	12167L Damage Level	10 W-us
Maximum DC Input	Low	10 VDC (On the RF Input)
	Standard	16 VDC (On the RF Input)
	12175:	N/A
Typical VSWR	12108L/12108S/12118L:	< 1.1:1
	12118S:	< 1.10:1
	12126L:	< 1.14:1
	12126S:	< 1.28:1
	12140L/12144L/12144S:	< 1.28:1 @ 40 GHz
	12150L:	< 1.39
	12150S:	< 1.39
	12154L:	<1.28:1 @ 44 GHz; <1.94:1 @ 54 GHz
	12167L:	<1.43:1 @ 67 GHz
	12175:	< 1.7:1
Typical Total RSS Error	12108L:	1.31%
	12108S:	1.09%
	12118L/12118S:	1.3% @ 15 GHz
	12126L:	1.71% @ 18 GHz
	12126S/12140L/12140S:	2.87% @ 30 GHz
	12144L/12144S:	2.77% @ 44 GHz
	12150L/12150S:	2.77% @ 50 GHz
	12154L:	2.77% @ 44 GHz; 3.73% @ 54 GHz
	12167L:	2.77%
	12175:	5.36%
Standard Connector	12108L/12108S/12118L/12118S:	High quality Type-N male
	12126L/12126S:	High quality 3.5 mm male
	12140L/12140S:	2.92 mm Male
	12144L/12144S:	High quality 2.4 mm male
	12150L:	High quality 2.4 mm male
	12150S:	High quality 2.4 mm male
	12154L:	High quality 2.4 mm male
	12167L:	High quality 1.85 mm male
	12175:	WR-15 waveguide
Recommended Cycle	1 year	

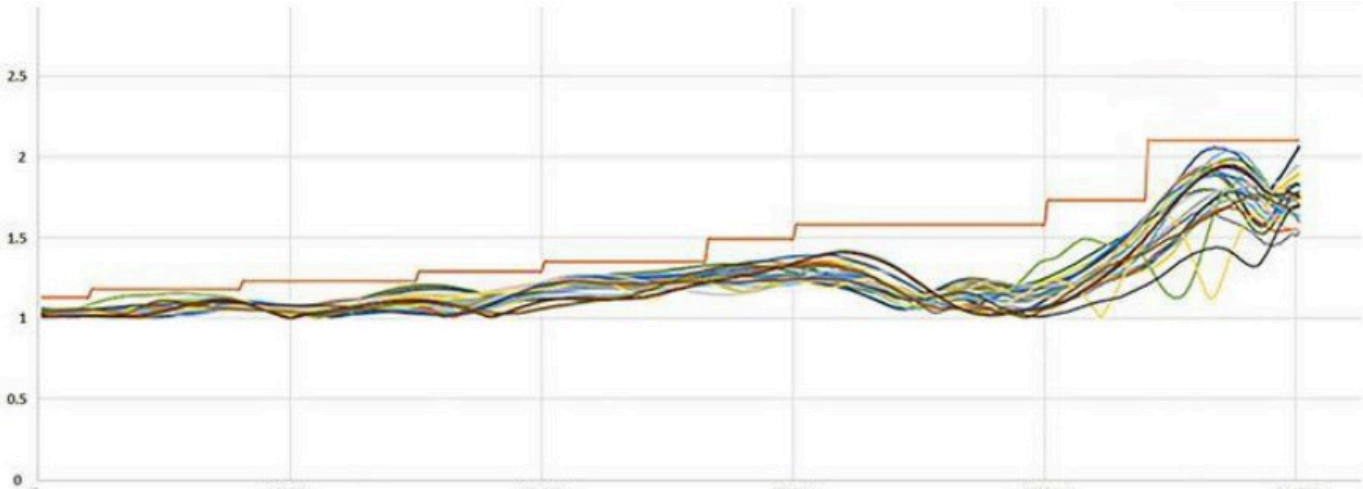
VSWR SPECIFICATIONS

12100L Series

Model	Band	VSWR
All 12100L	9 kHz-15 kHz	1.45
	15 kHz-40 kHz	1.25
	40 kHz-2 GHz	1.13
	2-8 GHz	1.18
12118L+	8-15 GHz	1.23
	15-18 GHz	1.29
12126L+	18-24.5 GHz	1.30
	24.5-26.5 GHz	1.35
12140L+	26.5-30 GHz	1.49
	30-40 GHz	1.58
12144L	40-44 GHz	1.73
12150L	44-50 GHz	1.39
12154L	50-54 GHz	2.20

12100S Series

Model	Band	VSWR
12108S+	1 MHz-2 GHz	1.13
	2-8 GHz	1.18
12118S	8-15 GHz	1.23
	15-18 GHz	1.29
12126S	18-24.5 GHz	1.30
	24.5-26.5 GHz	1.35
12140S+	15-20 GHz	1.29
	20-26.5 GHz	1.35
	26.5-30 GHz	1.49
	30-40 GHz	1.58
12144S	40-44 GHz	1.73
12150S	44-50 GHz	1.39
12167L	50-67 GHz	1.43
12175	50-75 GHz	1.70



12100L SERIES

Models / Connector	Band	Cal Factor Unc K=2
12108L-12140L Type-N	9-15 kHz	2.88%
	40 kHz-2 GHz	2.23%
	2-8 GHz	2.45%
12118L-12140L	10-18 GHz	2.85%
12126L/12140L	18-26.5 GHz	2.9%
12140L 2.92 mm	26.5-30 GHz	3.42%
	30-40 GHz	3.9%
	9 kHz-40 kHz	2.88%
12144L/12150L 2.4 mm	40 kHz-2 GHz	2.23%
	2-8 GHz	2.45%
	10-18 GHz	2.85%
	18-26.5 GHz	2.9%
	26.5-30 GHz	2.65%
	30-40 GHz	2.82%
	40-44 GHz	2.94%
	44-50 GHz	3.42%
	50-54 GHz	5.81%

Linearity Uncertainty

Input Level	Uncertainty
+10 to +20 dBm	3.0%
-10 to +10 dBm	2.0%
-20 to -10 dBm	3.5%
-60 to -20 dBm	2.0%

12100S SERIES

Models / Connector	Band	Cal Factor Unc K=2
12108S/12118S	5 MHz-2 GHz	1.79%
	2-8 GHz	1.53%
12118S	2-10 GHz	1.53%
	10-18 GHz	1.78%
12126S-12144S	5 MHz-2 GHz	2.23%
	2-10 GHz	2.45%
	10-18 GHz	2.85%
12140S	18-26.5 GHz	2.9%
	26.5-30 GHz	3.42%
	30-40 GHz	3.9%
12144S/12150S	26.5-30 GHz	2.65%
	30-40 GHz	2.82%
	40-44 GHz	2.94%

12167L - 1.85 mm male

Band	Cal Factor Unc K=2
9 kHz-40 kHz	2.88%
40 kHz-2 GHz	2.23%
2-10 GHz	2.45%
10-18 GHz	2.85%
18-26.5 GHz	2.9%
26.5-30 GHz	2.65%
30-40 GHz	2.82%
40-44 GHz	2.94%
44-50 GHz	3.42%
50-67 GHz	5.81%

12175 - WR-15 waveguide

Band	Cal Factor Unc K=2
50-75 GHz	5.78%

NOISE SPECIFICATIONS

Parameter	Specification	Typical
Average Mode -40 to +26 dBm	0.35%	0.025% to 0.15%
Average Mode -60 to -40 dBm	0.5 nW	0.2 nW
Normal Mode 2 ms Gate +10 to +26 dBm	0.35%	0.15%
0 to +10 dBm	0.5%	0.25%
-8 to 0 dBm	1.0%	0.5%
-25 to -8 dBm	1.5%	0.75%
-38 to -25 dBm	28 nW	20 nW
-45 to -38 dBm	95 nW	50 nW
Normal Mode 1 us Gate +10 to +26 dBm	0.45%	0.18%
0 to +10 dBm	3%	2%
-38 to 0 dBm	8.5%	1% to 4%

ZERO OFFSET SPECIFICATIONS

Model	Parameter	Specification
All 12100 Series	Zero Offset (Average Mode)	{[(1.0 nW @ 25 C) + Delta T x (0.075 nW/C)] +/- 0.01 nW/month}
12100S Series	Zero Offset (Normal Mode)	50 nW +/- 1 nW/month

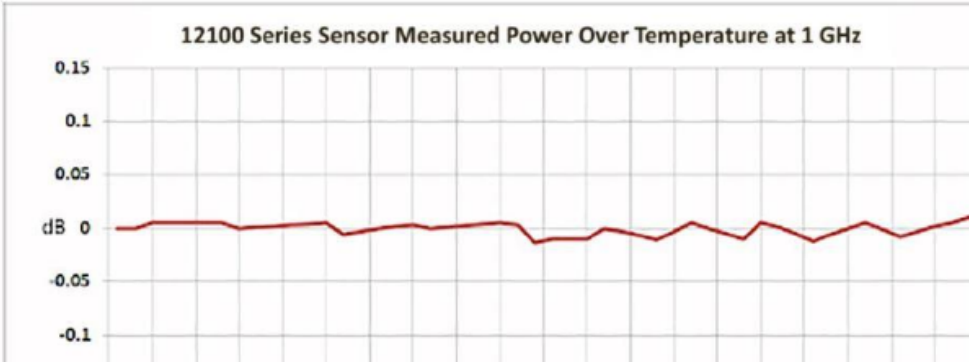
Model 12175 uses the applicable specifications within its -50 dBm to +18 dBm operating range.

TEMPERATURE SPECIFICATIONS

Model	Range	Specification	Typical
All 12100 Series	40 C-55 C	2.5%	0.5%
	30 C-40 C	0.5%	0%
	20 C-30 C	0%	0%
	10 C-20 C	0.5%	0%
	0 C-10 C	2.5%	0.5%

Thermal Stability

Berkeley Nucleonics’ thermal stability technology is utilized in the Model 12100 Series. Measurements remain stable over the entire operating temperature range. No user intervention, zeroing or calibration is required. This patented process also defines the zero-power conditions and eliminates zeroing requirements. Measurements are not interrupted for zeroing or calibration.



KEY FEATURES

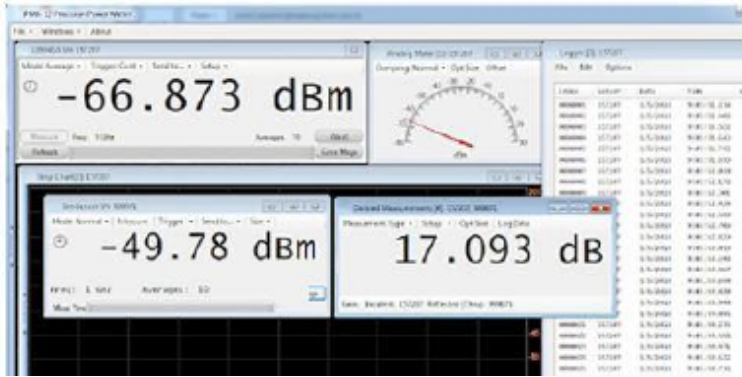
- RMS Responding & modulation independent
 - Interfaces: USBTMC and USBHID – Optional SPI or I2C TTL
 - Industry compatible IEEE 488.2 SCPI Command Set
 - Command compatible
 - Thermally Stable – no drift
- No user Zero required before use
 - Compatible with Windows, LINUX & More
 - Full dynamic range processed with each sample
 - Optional unattended autonomous capability & measurement storage
 - Security Options: MIL and SEC

BNC POWEREYE PRECISION METER SOFTWARE

The Model 12100L series sensors use BNC PowerEye Precision Power Meter software. The software allows users to make a wide variety of measurements from basic average power measurements to complex triggered measurements. Refer to the BNC PowerEye user's manual for further information.

In addition to BNC PowerEye, Berkeley Nucleonics provides an interactive IO utility with source code, a Persona utility, plus a variety of programmatic code and support.

BNC POWEREYE PRECISION POWER METER FEATURES



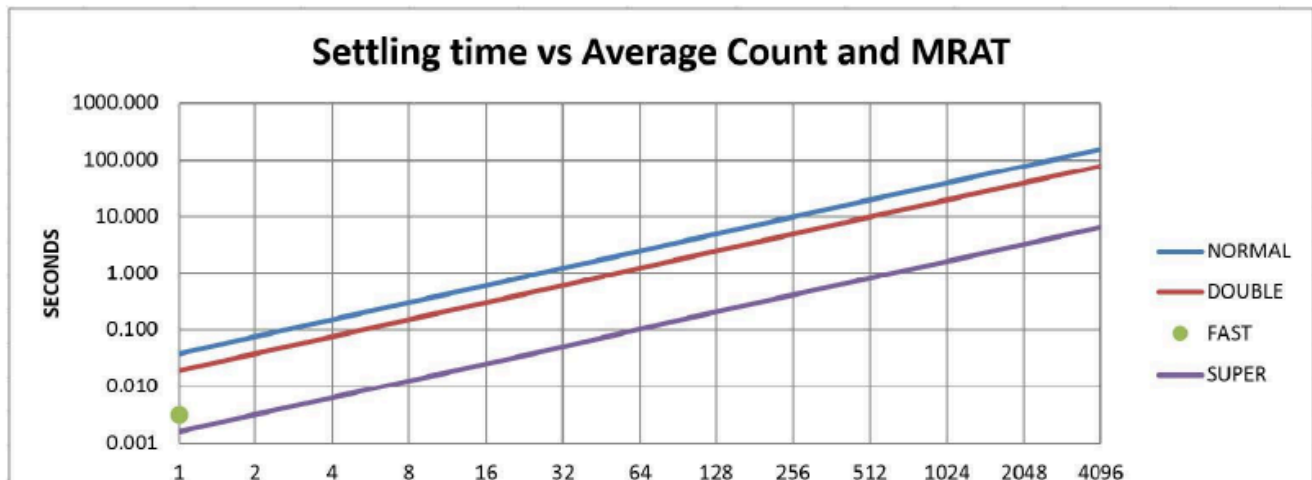
- For use with any 12100 Series Sensor
- Two sensor calculation windows with calculations for Gain, Loss VSWR, Reflection Coefficient, Mismatch Loss, Return Loss
- USB threading increases performance when using multiple sensors
- Triggering control including setting Level, Delay, Slope, Hysteresis, Impedance, more.
- Tabular logging with file storage and retrieval
- Controls Option UOP (Unattended Operation) and Option 001 (Analog Recorder Out)
- Recorder Out (analog out) control for scaling etc.
- Offset controls including simple offset and frequency dependent offset tables

All 12100 Series: Average Detector Mode Measurement Rate (MRAT)

Setting	Normal	Double	Fast	Super
Samples per measurement	384	192	32	16
Number of averages	1 to 1024	1 to 1024	1	1 to 4069
Measurement time per average	38.4 ms	19.2 ms	3.2 ms	1.6 ms

12100 Series: Normal Detector Mode Video Bandwidth

Parameter	Minimum	Typical
Auto Range	8 kHz	10 kHz
Averages per Range = 1	40 kHz	60 kHz
Range = 0	8 kHz	10 kHz



Interface Connectivity

The 12100 series sensors support both USBTMC and USB HID. Direct control of the sensor is also possible using optional SPI and I2C interface ports. These connectivity functions are optimized for programmatic control as well as unattended and logged power measurements. The sensors can be set up, controlled and data taken from any of the connectivity options. Berkeley Nucleonics' sensors are provided with a full featured application that utilizes the sensor's USB interfaces. Refer to the Option SPI documentation for information regarding I2C and SPI interfaces.

Interface	Detail
USBTMC	USB488 compliant
USB HID	USB Human Interface Device Class compliant
SPI / I2C (Option SPI)	Cable and connector included; cable may also power the sensor for unattended applications.

Remote Programming

The sensor is designed for full programmatic control in ATE systems and other applications requiring remote programming. 12100 Series sensors use SCPI (Standard Commands for Programmable Instruments) commands. The sensor's three connectivity options use the SCPI command set.

Parameter	Specification
Supporting Ports	USBTMC, USB HID, SPI/I2C
Command Set	SCPI
Compatibility	USBTMC programmatic control, NI and Agilent VISA IO libraries, Agilent U2000, and other SCPI systems.

Analog Recorder Output

Option 001, Analog Recorder Out. Recorder Output is a filtered analog output that can be used for various purposes. The output provides an accurate scaled voltage representing the power level of the signal. The output is linear (not log dB) and can be scaled. When enabled, Recorder Out uses the same SMB port used by Trigger Out making the port unavailable for triggering use. Trigger in is not affected.

Parameter	Specification
Output filter bandwidth	Average Detector: 0.001 Hz to 32 Hz; Normal Detector: 0.001 Hz to 81 Hz
Output range	0 to 1 V into 1,000 ohms
Output impedance	1 kOhm
Resolution	+/- 25 uV
DC Offset	0 to 5 mV
Scale	Linear
Connector	SMB Male

Triggering

Parameter	Specification
Trigger source	Internal, Immediate, External
Trigger Delay range	+/- 10 s
Resolution	1 us
Trigger Out pulse width	500 ns

Internal Triggering

Parameter	Specification
Level	Approximately -50 dBm to 20 dBm
Level resolution	0.1 dB
Slope	Positive or Negative
Hysteresis	0 dB to 3 dB

External Triggering

Parameter	Specification
Trigger Input	Min high 2.0 V; max low 0.8 V
Input Load	Selectable 100 kOhm or 50 Ohm
Timing requirements	Minimum width 25 ns on / 25 ns off; repetition 50 ns minimum
Absolute input limits	+5.5 V max; -0.5 V min

Store, Recall and Logging Memory

The 12100 series contains volatile and non-volatile memory.

Store and recall functions for the sensor's state and register functions such as Frequency, Averages and Analog Recorder Out settings have a lifetime of 1 million write and erase cycles.

When Option UOP is present, the sensor contains separate, non-volatile flash memory that is designed for long term logging of measurements.

Memory & Storage Specifications

Parameter	Specification
Non-volatile NAND flash	50 million measurements
Maximum storage rate	1000 measurements per second

Unattended Operation (UOP)

Unattended operation is used to make autonomous measurements. Once set up using a computer, the sensor only requires power to function. Measurements are stored in the sensor's non-volatile memory and are time stamped using the sensor's internal real time clock. The option allows the use of trigger functions. Recorder output can be enabled while in unattended operation allowing calibrated analog output functions with no computer or power meter connected. Power can be applied using a USB power only cable or if Option SPI has been purchased, its ribbon cable can be used to apply power. Measurement storage using UOP is not possible when Option MIL is installed.

Real Time Clock

Parameter	Specification
Time Accuracy	Typical: unpowered 20 ppm at 25 C; under power and stable, 2 ppm
Functionality	Timestamp data for measurements stored in memory
Backup	Super cap; 1 day typical RTC operation when fully charged

Environmental	Operating	Storage
Temperature	0 C to 55 C	-25 C to 85 C
Humidity	15% to 95% non-condensing	15% to 95% non-condensing
Altitude	10,000 ft (3,000 m)	50,000 ft (15,000 m)

Initial Stabilization Time

For general use, the 12100 series sensors are stable 5 minutes after electrical power is applied. Specifications detailed in this datasheet are valid ONLY after a 30 minute warm up period and for continuous wave (CW) signals unless otherwise stated. The recommended calibration interval for this product is one year. Specifications apply over the listed temperature and relative humidity range unless otherwise stated.

Option MIL

Option MIL is designed to address security and data sanitization issues. When purchased with this option, the user cannot write to any non-volatile memory. Consult the factory for additional information regarding option MIL.

Sanitization Option (SEC)

The sanitization option adds secure erase capability. When the SEC command is executed, all non-volatile memory is erased including User Presets, Persona information, Store & Recall data, User Cal, Simple offsets and FDO tables. The process is an erase - random overwrite erase process. For additional security users can execute the command repeatedly. The option is not available if Option MIL is purchased because non-volatile memory writes are disallowed with the option.

MODEL 12100 SERIES: ORDERING OPTIONS

Specify the model number, which sets the frequency reach and standard connector. Choose S builds for coverage from 1 MHz, L builds for coverage from 9 kHz, or Model 12175 for 50 GHz to 75 GHz with WR-15 waveguide.



ATE Mounting Bracket
(Also shown with SPI
cable)



Secure USB Cable



SPI / I²C
Demonstration Kit Available



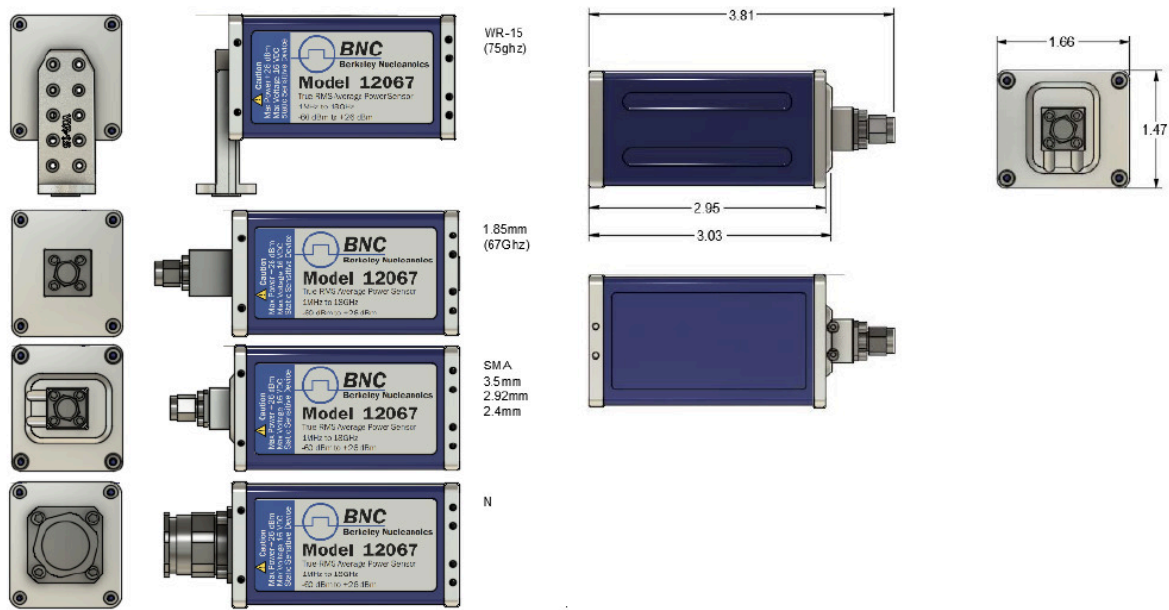
Optional Connectors
Available

Model	Standard Connector
12108S/L, 12118S/L	Type-N male
12126S/L	3.5 mm male
12140S/L	2.92 mm male
12144S/L, 12150S/L, 12154L	2.4 mm male
12167L	1.85 mm male
12175	WR-15 waveguide

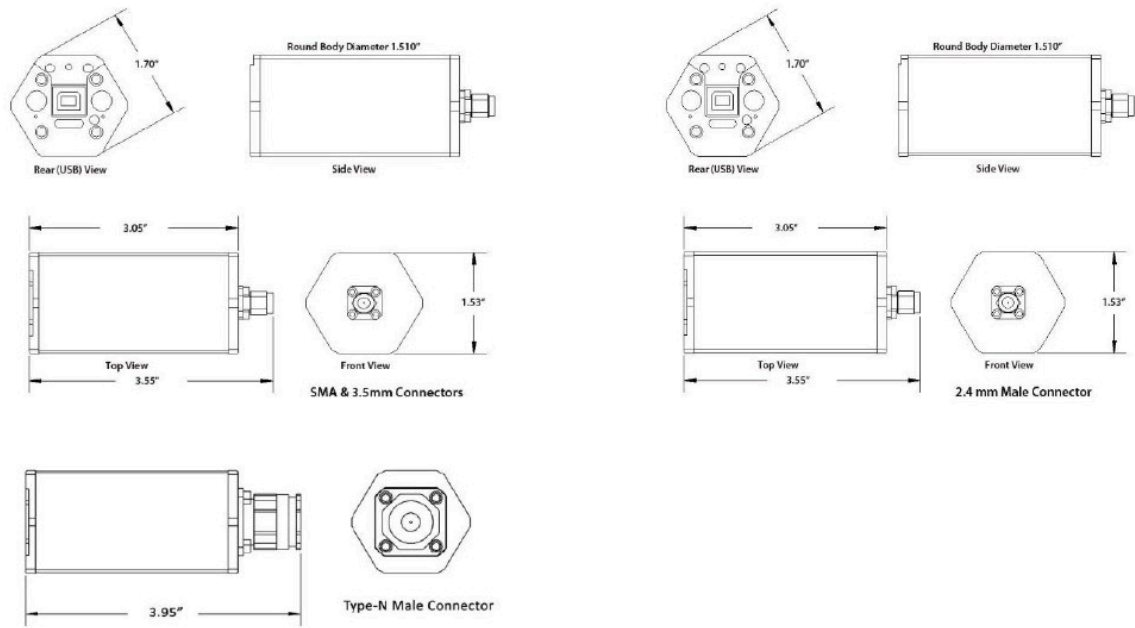
MODEL 12100 SERIES OUTLINE DRAWINGS

New Chassis

New Chassis



Dimension tolerance +/- 0.063"



Dimension tolerance +/- 0.063"